

Anika Tabassum

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RESEARCH INTEREST

Air Quality · GIS & Remote Sensing · Climate-Resilient Planning · Environmental Data Analysis

EDUCATION

Khulna University of Engineering & Technology (KUET) Jan 2020 - Nov 2025
Bachelor of Urban and Regional Planning (BURP) | CGPA: 3.55 / 4.00

RESEARCH PROJECTS

Assessing the Temporal Dynamics of PM_{2.5} and PM₁₀ in Khulna City with Meteorological and Trajectory-based Analysis (*Undergraduate Thesis*) Nov 2024 - Sept 2025

Examined how PM_{2.5} and PM₁₀ levels in Khulna changed through 2024 using hourly data from the Department of Environment CAMS-8 station and NOAA-HYSPLIT trajectories, with Pearson correlation, multiple linear regression, K-Means clustering, and STL decomposition. Found winter and post-monsoon peaks of about 155 µg/m³ for PM_{2.5} and 220 µg/m³ for PM₁₀ against 15-40 µg/m³ in the monsoon, with regression models reaching R² of 0.79-0.91.

Meteorological and HYSPLIT-Derived Transport Influences on PM_{2.5} Exposure in a Khulna City, Bangladesh (*Co-authored manuscript from thesis; under editorial assessment*) Nov 2025 - Mar 2026

Assessed how weather and regional air transport influence PM_{2.5} variability in Khulna using hourly 2024 monitoring data, NASA POWER and NCEI meteorological data, and NOAA-HYSPLIT backward trajectories, with regression, Random Forest, K-Means clustering, and STL decomposition in Python. Found winter PM_{2.5} (~121 µg/m³) nearly eight times the WHO daily guideline against ~15 µg/m³ in the monsoon, with Random Forest explaining 64% of daily variance.

Spatial and Temporal Analysis of Air Quality Using Remote Sensing and GIS in Fakirhat Upazila (*CEGIS Internship Research*) Oct 2024

Mapped air quality across Fakirhat Upazila using Landsat 8 imagery (2024), ArcGIS, IDW interpolation, and AQI mapping, extending data from only three ground stations. Found PM_{2.5} was the only pollutant in the moderate AQI category, while vegetation (38.9%) and water bodies (28.7%) appeared to buffer pollution, and surface temperature had a weak effect (R² < 0.04 for most).

WORK EXPERIENCE

Rupantar - Young Professional Sep 2025 - Sep 2026
Conducted community vulnerability assessments and supported local development planning under the GO4IMPact project (WaterAid & Swisscontact) in Satkhira. Supported 80+ Participatory Wellbeing and Vulnerability Assessment (PWVA) reports, helped prepare Union Parishad Five-Year Plans, and conducted household surveys and community consultations using mWater and ArcGIS.

CEGIS, Bangladesh - Intern, Ecology, Forestry & Biodiversity Division Oct 2024
Contributed to an air quality assessment and geospatial analysis project by applying statistical analysis and presenting findings to project stakeholders. I used ArcGIS, Google Earth Engine, and USGS Earth Explorer.

TECHNICAL SKILLS

Geospatial & Remote Sensing: ArcGIS, Google Earth Engine, USGS Earth Explorer, NOAA-HYSPLIT Model

Programming Languages: Python, R

Data & Statistics: Microsoft Excel, SPSS

Survey & Data Collection: mWater, KoboToolbox

Other Essentials: Microsoft 365, AutoCAD, Adobe Illustrator

TRAINING & PARTICIPATION

Union Parishad Governance and Climate-Resilient Integrated Water Resource and Solid Waste Management Jul - Aug 2026

Completed a three-phase, 3-day training organized by NILG under WaterAid Bangladesh-supported initiatives, covering local governance and climate-resilient water and solid waste management, and prepared formal reports documenting its proceedings.

Five-Year Planning Training Jun 2026

Completed the WaterAid Bangladesh-supported training and prepared a formal report on its key outcomes.

Nature-Based Future Challenge - Team Delta Avatars 2024

Southwestern Bangladesh's delta region needs climate-resilient strategies and stronger water governance to adapt to climate change. As part of Team Delta Avatars, I advanced to Round 2 by contributing spatial analysis and geospatial detailing of the region's water systems in ArcMap, supported by a literature review and area analysis on sustainable water governance and climate adaptation.

GenU Bangladesh imaGen Ventures Youth Challenge on Climate Change Impact - Team Coastal Watch 2022

Bangladesh's coastal areas are highly exposed to climate change, including salinity that limits access to safe water, and need stronger resilience. With Team Coastal Watch, I advanced to the regional round by proposing and demonstrating a rooftop water-harvesting system with charcoal-based filtration to provide safer water in salinity-affected coastal communities.

AWARDS & ACHIEVEMENTS

Dean's List (Faculty of Civil Engineering, KUET) Academic Session 2019-2020

Named to the Dean's List for outstanding academic performance in the first year, achieving a GPA of 3.79 out of 4.00.

2nd Runner-Up - Vision in Motion (Team Durbeen) Plannation 3.0

Secured third place in the Vision in Motion competition at Plannation 3.0, the Department of Urban and Regional Planning's annual flagship event.

LEADERSHIP & EVENT MANAGEMENT

Head, Magazine Publication - Plannation 3.0 Feb 2025

Led content creation for Plannation 3.0, the Department of Urban and Regional Planning's flagship annual event, in the KUET event magazine, and hosted two event segments, contributing to smooth execution.

Assistant Secretary of Media - KUET Business and Entrepreneurship Club Mar 2023 - Apr 2024

Served on the media team of the KUET Business and Entrepreneurship Club (KBEC), after joining the club as a Junior Executive in April 2022.

Assistant Director of Communications - Hult Prize, KUET Dec 2022 - Feb 2023

Supported communications activities for Hult Prize KUET, the university chapter of the international student entrepreneurship competition, after serving as an Organizing Committee member for the 2022 edition.

Campus Ambassador - Roboadda (Tech Hub) Jun 2021 - Dec 2021

I led digital outreach and content strategy using Canva, driving brand engagement for the organization.

EXTRA-CURRICULAR ACTIVITIES

Beyond academics, I have remained actively engaged in entrepreneurship and the performing arts. I founded and ran Bakehaus, a personal baking business, gaining practical experience in managing a small venture and interacting with customers. I also enjoy dance and have participated in an inter-university dance fest, reflecting my continued involvement in creative and extracurricular activities.

LANGUAGE PROFICIENCY

Bangla: Native | English: IELTS Academic overall 7.0 (Listening 7.5, Reading 6.5, Writing 6.5, Speaking 7.0)